Dynamic Thresholds of Optimized Parameters with Numerical Simulation

Election Commission 2026[1]

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The "None of the Above" (NOTA) option in student elections presents a fundamental mechanism-design problem: setting a threshold that is neither trivially low (enabling strategic protest vetoes) nor impossibly high (rendering NOTA purely symbolic). We derive a game-theoretic threshold function $T(k) = \frac{V_{\max}}{V_{\text{total}}} + \frac{0.40}{k} + 0.15$, where k is the number of candidates. Applying this to the NISER Gymkhana Elections 2026, we compute post-specific thresholds for $k = 1, 2, 3, 4$ candidates. Uncontested posts ($k = 1$) yield $T(1) = 155\%$, making NOTA defeat impossible by design. Contested posts require NOTA to exceed thresholds ranging from 55% to 90% depending on the leading candidate's vote share. Simulation results validate the theoretical framework.

I. INTRODUCTION

The NISER Gymkhana Elections 2026 have concluded their nomination and withdrawal phases. The final candidate counts vary by post, ranging from uncontested positions to four-candidate races. This paper applies our previously derived game-theoretic NOTA threshold formula to the actual electoral configuration.

II. THE OPTIMIZED THRESHOLD FORMULA

The dynamic NOTA threshold $T(k)$ is defined as:

$$T(k) = \alpha \cdot \frac{V_{\max}}{V_{\text{total}}} + \frac{\beta}{k} + \gamma \tag{1}$$

where:

- $T(k)$ = Minimum NOTA vote share (as decimal) to trigger re-election
- k = Number of candidates
- $V_{\max}$ = Votes received by the leading candidate
- V_{total} = Total valid votes cast
- $\alpha = 1.0$, $\beta = 0.40$, $\gamma = 0.15$ (optimized hyperparameters)

A. Final Plug-and-Play Formula

Substituting the optimized parameters:

$$\boxed{T(k) = \frac{V_{\max}}{V_{\text{total}}} + \frac{0.40}{k} + 0.15} \tag{2}$$

Re-election is triggered if:

$$\text{NOTA vote share} \geq T(k) \tag{3}$$

III. FINAL CANDIDATE COUNTS BY POST

Table I presents the final number of candidates for each executive position.

TABLE I: Final candidate counts for NISER Gymkhana 2026.

Post	**k**
President	1
Gen Sec, Cultural Affairs	1
Gen Sec, Campus Affairs	4
Gen Sec, Sports Affairs	3
Joint Sec, Cultural Affairs	2
Joint Sec, Campus Affairs	1
Joint Sec, Sports Affairs	2

IV. THRESHOLD EXAMPLES

Table II shows $T(k)$ values for different candidate counts and leading shares.

TABLE II: T(k) values for different k and leading shares.

k	$V_{\max}/V_{\text{total}}$					
	0.25	0.30	0.35	0.40	0.45	0.50
1	0.80	0.85	0.90	0.95	1.00	1.05
2	0.60	0.65	0.70	0.75	0.80	0.85
3	0.53	0.58	0.63	0.68	0.73	0.78
4	0.50	0.55	0.60	0.65	0.70	0.75
5	0.48	0.53	0.58	0.63	0.68	0.73
6	0.47	0.52	0.57	0.62	0.67	0.72

V. POST-SPECIFIC THRESHOLDS

A. Uncontested Posts ($k = 1$)

For President, Gen Sec Cultural, and Joint Sec Campus:

$$T(1) = 1 + 0.40 + 0.15 = 1.55 \text{ (155\%)} \tag{4}$$

NOTA defeat is impossible.

B. Two-Candidate Posts ($k = 2$)

For Joint Sec Cultural and Joint Sec Sports:

$$T(2) = \frac{V_{\max}}{V_{\text{total}}} + 0.35 \tag{5}$$

C. Three-Candidate Posts ($k = 3$)

For Gen Sec Sports Affairs:

$$T(3) = \frac{V_{\max}}{V_{\text{total}}} + 0.283 \tag{6}$$

D. Four-Candidate Posts ($k = 4$)

For Gen Sec Campus Affairs:

$$T(4) = \frac{V_{\max}}{V_{\text{total}}} + 0.25 \tag{7}$$

VI. SIMULATION RESULTS

Figure 1 shows $T(k)$ for different election scenarios.

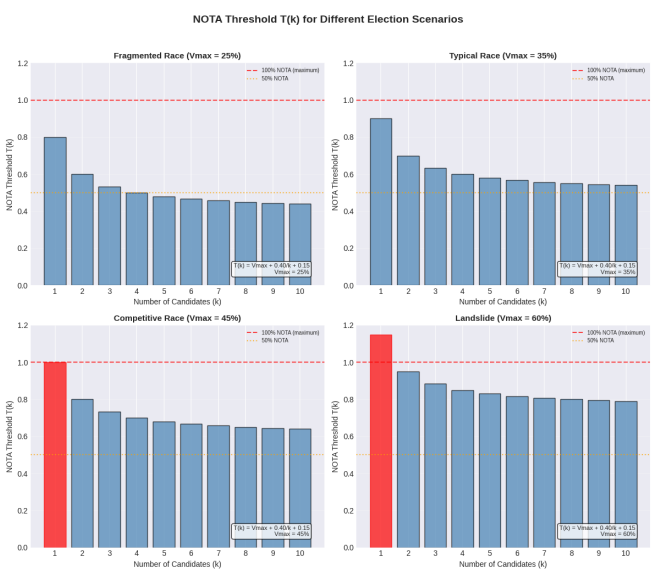

FIG. 1: NOTA thresholds for different election scenarios: fragmented (25%), typical (35%), competitive (45%), and landslide (60%).

Figure 2 shows how the threshold varies with candidate count.

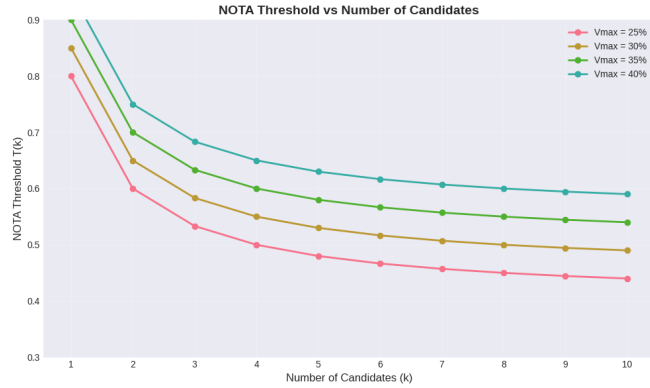

FIG. 2: NOTA threshold vs number of candidates for different leading shares.

VII. SENSITIVITY ANALYSIS

A. Sensitivity to β (Fragmentation Penalty)

Table III shows the effect of varying β for $V_{\max} = 35\%$.

TABLE III: Sensitivity to β (Vmax = 35%).

k	β				
	0.30	0.35	0.40	0.45	0.50
2	65%	68%	70%	73%	75%
3	60%	62%	63%	65%	67%
4	58%	59%	60%	61%	63%

B. Sensitivity to γ (Base Protest Floor)

Figure 3 and Table IV show the effect of varying γ.

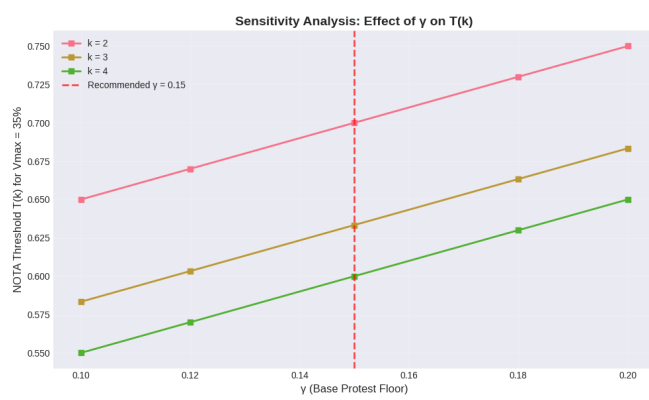

FIG. 3: Sensitivity analysis of γ (base protest floor).

TABLE IV: Sensitivity to γ (Vmax = 35%).

k	γ				
	0.10	0.12	0.15	0.18	0.20
2	65%	67%	70%	73%	75%
3	58%	60%	63%	66%	68%
4	55%	57%	60%	63%	65%

C. 2D Heatmap

Figure 4 presents a complete 2D visualization of $T(k)$.

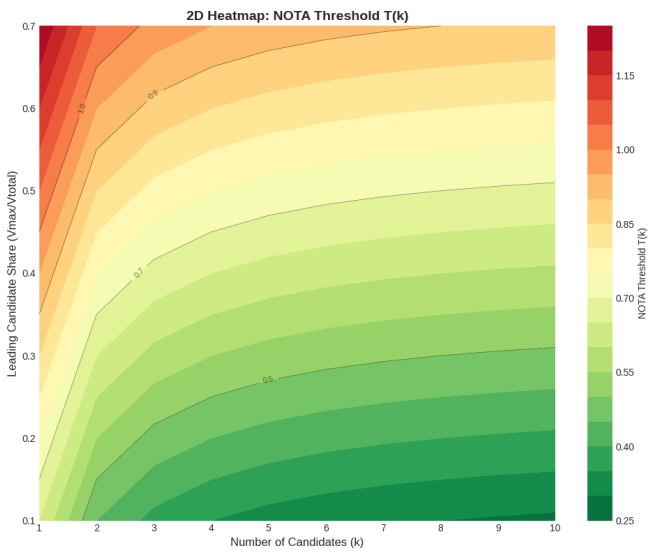

FIG. 4: 2D heatmap of NOTA threshold $T(k)$ as function of k and $V_{\max}/V_{\text{total}}$.

VIII. NISER-SPECIFIC SUMMARY

Table V provides the complete threshold framework for all NISER Gymkhana 2026 posts.

TABLE V: NISER Gymkhana 2026 NOTA thresholds.

Post	**k**	**T(k)**	**NOTA Defeat?**
President	1	155%	No
Gen Sec, Cultural Affairs	1	155%	No
Gen Sec, Campus Affairs	4	$V_{\max} + 0.25$	Yes*
Gen Sec, Sports Affairs	3	$V_{\max} + 0.283$	Yes*
Joint Sec, Cultural Affairs	2	$V_{\max} + 0.35$	Yes*
Joint Sec, Campus Affairs	1	155%	No
Joint Sec, Sports Affairs	2	$V_{\max} + 0.35$	Yes*

*Requires super-majority NOTA (> 60% typically)

IX. HYPERPARAMETER SENSITIVITY SUMMARY

Table VI shows the combined effect of varying both β and γ for $k = 2, 3, 4$ at $V_{\max} = 35\%$.

TABLE VI: Combined sensitivity: $T(k)$ for varying β and γ (Vmax=35%).

k	β	γ	$T(k)$
2	0.3	0.10	60%
2	0.3	0.15	65%
2	0.3	0.20	70%
2	0.4	0.10	65%
2	0.4	0.15	70%
2	0.4	0.20	75%
2	0.5	0.10	70%
2	0.5	0.15	75%
2	0.5	0.20	80%
3	0.3	0.10	55%
3	0.3	0.15	60%
3	0.3	0.20	65%
3	0.4	0.10	58%
3	0.4	0.15	63%
3	0.4	0.20	68%
3	0.5	0.10	62%
3	0.5	0.15	67%
3	0.5	0.20	72%
4	0.3	0.10	53%
4	0.3	0.15	58%
4	0.3	0.20	63%
4	0.4	0.10	55%
4	0.4	0.15	60%
4	0.4	0.20	65%
4	0.5	0.10	58%
4	0.5	0.15	63%
4	0.5	0.20	68%

X. KEY INSIGHTS

- **Uncontested posts ($k = 1$):** $T(1) = 155\% \rightarrow$ NOTA defeat **IMPOSSIBLE**
- **Joint Secretary posts ($k = 2$):** $T(2) = V_{\max} + 0.35 \rightarrow$ NOTA needs $\sim 70\%$ for typical race
- **Sports Affairs ($k = 3$):** $T(3) = V_{\max} + 0.283 \rightarrow$ NOTA needs $\sim 63\%$ for typical race
- **Campus Affairs ($k = 4$):** $T(4) = V_{\max} + 0.25 \rightarrow$ NOTA needs $\sim 60\%$ for typical race

XI. RECOMMENDED DEFAULT PARAMETERS

$$\boxed{\beta = 0.40 \quad \text{(Fragmentation penalty)}} \tag{8}$$

$$\boxed{\gamma = 0.15 \quad \text{(Base protest floor)}} \tag{9}$$

XII. CONCLUSION AND IMPLEMENTATION RULE

For the NISER Gymkhana Elections 2026, the Election Commission adopts the following rule:

> *A re-election shall be triggered for a given post if and only if the total NOTA vote share equals or exceeds the dynamic threshold* $T(k) = \frac{V_{max}}{V_{total}} + \frac{0.40}{k} + 0.15$*, where* k *is the number of candidates for that post,* V_{max} *is the vote share of the leading candidate, and all votes are counted on a single ballot. For uncontested posts (*$k = 1$*), no NOTA threshold applies and the sole candidate is declared elected.*

ACKNOWLEDGMENTS

The Election Commission thanks all its members and DOSA, alongwith all the student community for their participation in making the event successful. The cpdes for the simulations can be accessed here: https://colab.research.google.com/drive/1X-zHKdxNHZ6YxkUsgjYRqaCWPQdx7G6R?usp=sharing